Genetic Algorithm for the Travelling Salesman Problem

A study on 15 Indian cities, validated against the exact optimum

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Submitted to: Dr. Neha Chaudhary

Academic Year 2026–27

Outline

1. Problem and Approach
2. GA Design
3. Results
4. Conclusion

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The Travelling Salesman Problem

Visit each of 15 Indian cities exactly once and return to the start, minimising total distance.

$$L(\pi) = \sum_{i=1}^{n} d\big(\pi_i, \pi_{(i \bmod n)+1}\big)$$

- $(15-1)!/2 \approx 4.4 \times 10^{10}$ distinct tours
- NP-hard, so brute force is impractical
- Distances: great-circle (haversine) between city coordinates

Why a Genetic Algorithm?

- Population-based search inspired by natural selection
- Explores many regions of the search space in parallel
- Recombination combines good partial routes from different tours

Objectives

Implement a GA from scratch, validate it against the exact optimum (Held–Karp), compare with baselines, and study mutation rate and population size.

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GA Design

Component	**Choice**
Chromosome	Permutation of city indices
Fitness	$1/L(\pi)$
Selection	Tournament, $k = 5$
Crossover	Order Crossover (OX1), prob. 0.9
Mutation	Segment inversion, prob. 0.2
Elitism	Best 2 copied unchanged
Population / generations	150 / 300

A permutation encoding means every chromosome is a valid tour.

Worked Example: Order Crossover

Parents (8 cities), slice at positions 2–4:

- P1 = [1, 2, 3, 4, 5, 6, 7, 8]
- P2 = [3, 7, 5, 1, 6, 8, 2, 4]

Keep [3, 4, 5] from P1, fill the rest in P2 order: 7, 1, 6, 8, 2.

Child

[7, 1, 3, 4, 5, 6, 8, 2]

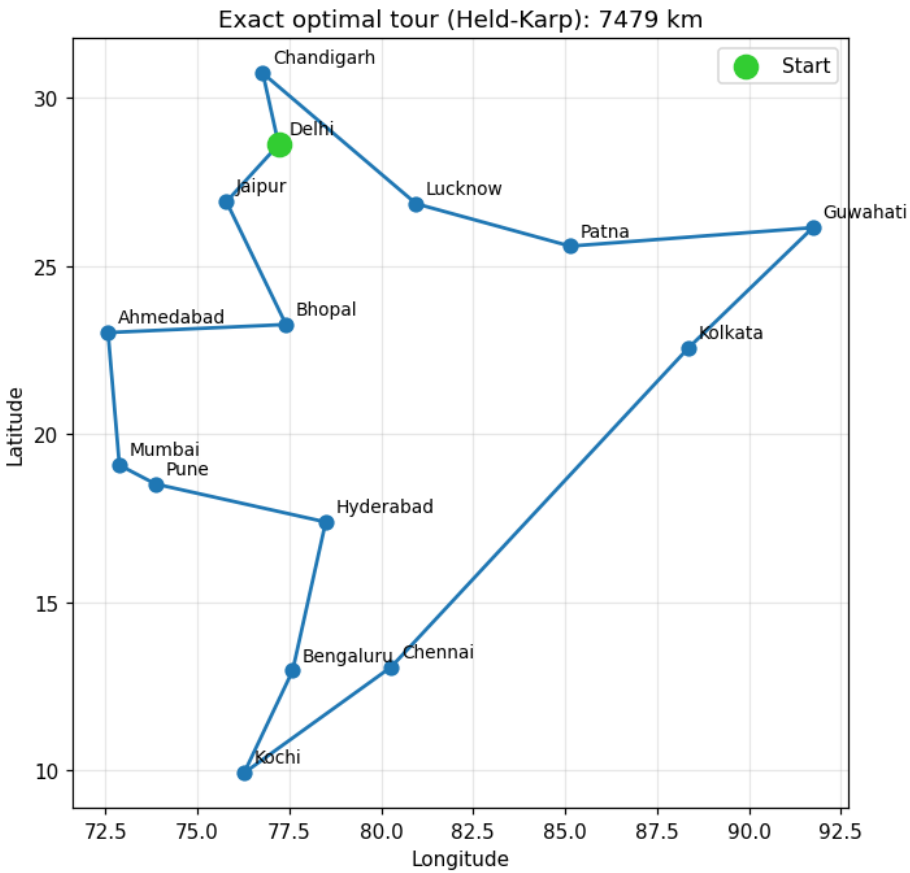
Inversion mutation

[1, 2, 3, 4, 5, 6, 7, 8] → [1, 2, 6, 5, 4, 3, 7, 8]

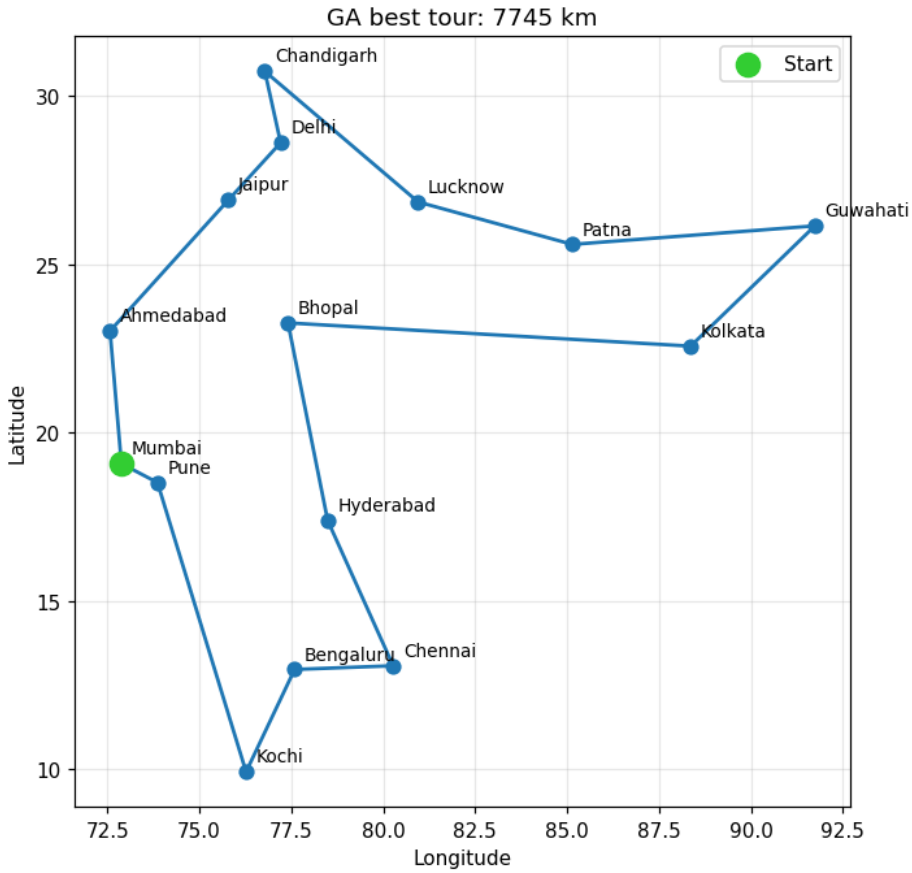
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Exact Optimum vs. GA Tour

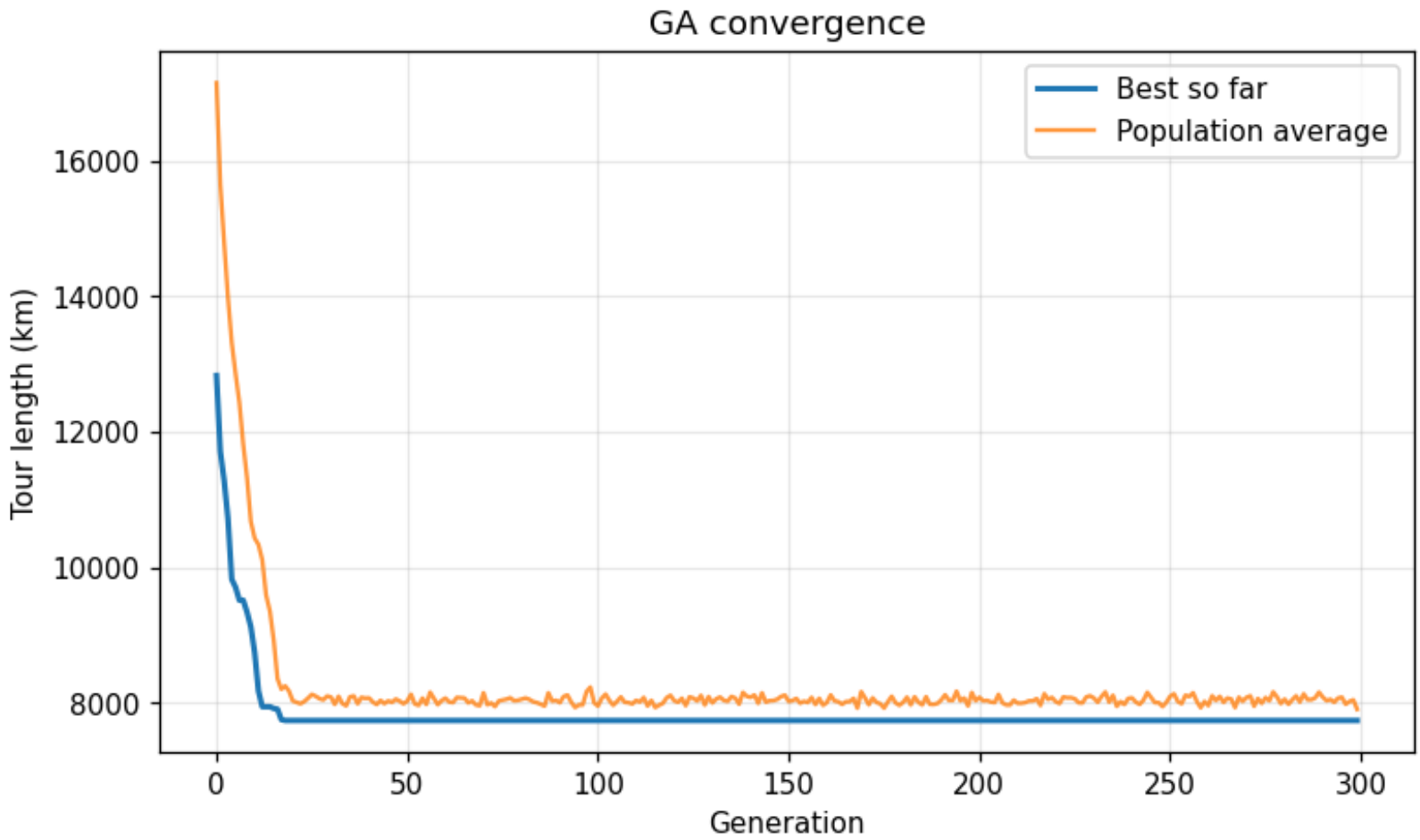


Held–Karp: 7478.9 km



GA, one run (seed 42): 7745.4 km

Convergence



The best tour improves quickly while the population average follows.

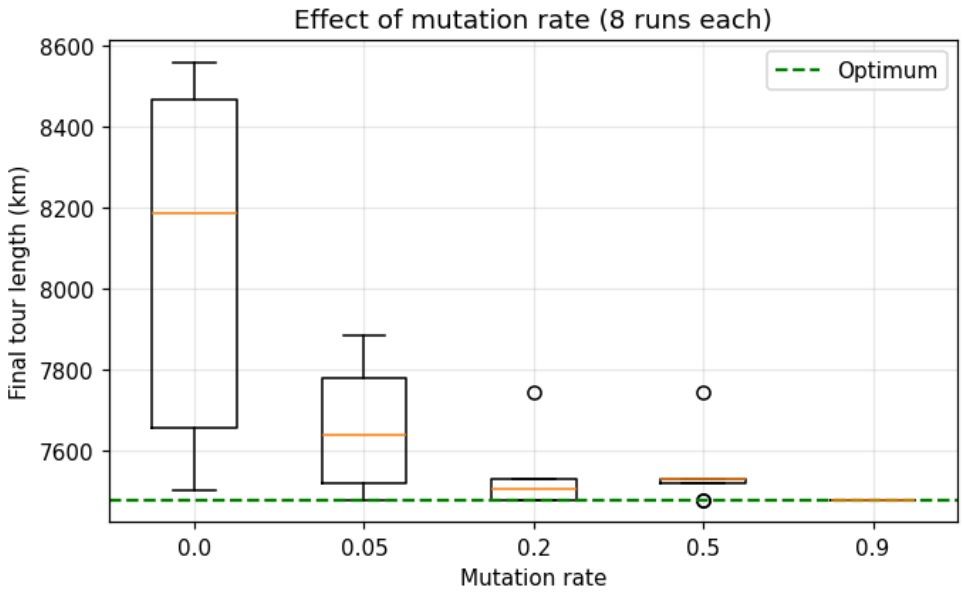
GA versus Baselines

Method	Length (km)	Gap (%)
Random search (same budget)	9930.4	32.8
Nearest neighbour	9169.4	22.6
GA (mean of 10 runs)	7526.9	0.6
GA (best of 10 runs)	7478.9	0.0
Exact optimum (Held–Karp)	7478.9	0.0

Key result

The GA found the true optimum in 5 of 10 runs, and averaged 0.6% above optimal.

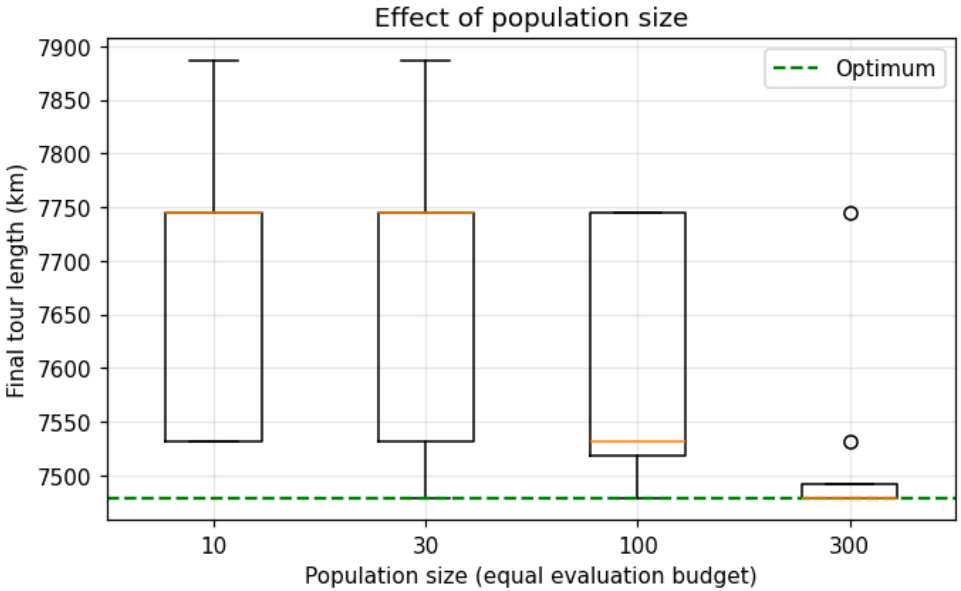
Effect of Mutation Rate



Rate	Mean (km)
0.00	8079.0
0.05	7660.8
0.20	7532.2
0.50	7545.6
0.90	7478.9

No mutation performs worst: it stagnates.

Effect of Population Size



Pop.	Gens	Mean (km)
10	4500	7683.1
30	1500	7676.5
100	450	7598.8
300	150	7518.9

Equal budget of 45,000 evaluations.

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Complexity and Limitations

- GA time $O(G \cdot P \cdot n)$, space $O(P \cdot n)$
- Held–Karp: $O(n^2 2^n)$ time, $O(n\,2^n)$ space, impractical beyond about 25 cities
- No optimality guarantee; results depend on parameters and seed
- Premature convergence when diversity is lost (seed 42 ended 3.6% above optimal)

Conclusion

- Within 0.6% of the exact optimum on average, evaluating only 45,000 of 4.4×10^{10} tours
- Clearly beats random search (32.8%) and nearest neighbour (22.6%)
- Mutation prevents stagnation; larger populations win at a fixed budget
- Repeated runs are needed because a single run can get stuck

Thank you! Questions?